

RODEO - WP4: Robotics-guided spine surgery Dataset for visual perception CT-scan vertebral and intra-vertebral annotation

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Abstract In the context of scoliosis treatment surgery, several CT-scans are acquired throughout the patient's journey: clinical diagnosis and surgery planning (meaning planning of pedicle screw placement), per-operative planning for pedicle screw insertion with navigation, and lastly verification of the surgery. In the context of the RODEO ANR project, we aim to supplement the CT-scan with the segmentation of vertebrae and its anatomy to help the surgeon in its task and to constitute a visual perception dataset for robotic planning. The dataset RODEO contains 48 scoliotic CT-scans, collected at APHP Armand Trousseau (Paris, France) with vertebrae-level and sub-vertebrae-level annotations, and will be expanded to more CT-scans in the future. This dataset is the first scoliosis-focused dataset and the first sub-vertebrae annotation dataset. We believe that this dataset will facilitate further research in many spine-related image analysis tasks, especially scoliosis-related.

Code is available at <https://git.isir.upmc.fr/maxellende/Segmentation>.

Checkpoints and results are available at https://git.isir.upmc.fr/maxellende/Segmentation_data_results_checkpoints.

Project cloud is at this link <https://cloud.isir.upmc.fr/apps/files/files/34703852?dir=/RODEO>.

Slides summarizing project launch and WP4 for 2025 are available at https://cloud.isir.upmc.fr/apps/files/files/38919163?dir=/RODEO/R%C3%A9union%20de%20retrait%C3%A9%202025_10_09&editing=false&openfile=true.

Contents

1	Introduction	3
1.1	Context	3
2	Data annotation	4
2.1	Overview of scientific literature	4
2.1.1	Vertebrae annotation	4
2.1.2	Intra-vertebrae annotation	6
2.2	Workflow	6
2.3	Results	6
2.3.1	Experimental set-up	7
2.3.2	Metrics	7
2.3.3	Quantitative results	7
2.3.4	Qualitative results	7
3	Conclusion & Perspectives	8
A	Vertebrae datasets	10
A.1	Vertebrae datasets illustrations	10
A.2	Vertebrae datasets comparison	11
B	nnU-Net for vetebrae segmentation: parameters and cross-validation results	11
B.1	nnU-Net configuration parameters	11
B.2	Best configuration of nnU-Net using cross-validation	11

1 Introduction

1.1 Context

The RODEO ANR project is set to reshape surgical robotics by adapting cutting-edge AI, targeting the challenging spine surgery. The project is conducted within the Machine Learning for Robotics (MLR) team at the robotics lab Institut des Systèmes Intelligents et de Robotique (ISIR), Sorbonne University (SU). The application framework is robot-guided surgery for the spine, based on a surgical platform available at ISIR.

In this context, a 7-degree-of-freedom (DoF) robotic arm is equipped with various sensors (position, speed, force, electrical conductivity, vibrations) and used during surgical interventions, such as the insertion of pedicle screws into the spine. This current robotic platform uses a set of already implemented control laws (e.g., position, speed, force control) to execute surgical tasks or sub-tasks, such as drilling a preliminary trajectory for pedicle screw placement. Before the operation, a 3D scan of the patient is performed, allowing the surgeon to define the medical procedure to be followed during surgery.

A dataset for visual perception. The precise localization of the robot during intervention will be based on segmenting medical structures on 3D-reconstructed scenes from RGB, depth or LiDAR images flow of the surgical scene and registering them with respect to pre-operative 3D CT-scans. The dataset will thus be used for robotic perception and planning and as well in the evaluation of the following work packages of the project: sustainable multi-modal planning (WP1), residual hybrid controllers (WP2) and uncertainty quantification (WP3).

First of its kind dataset It exists many datasets of annotated CT-scan of vertebrae, but none containing a significant number of cases of scoliosis. Also, there is no currently available dataset with annotations at a sub-vertebral level, meaning annotations of pedicles, vertebrae body, articular processes, etc. This is why this dataset collection is such a challenge and would be beneficial for the community.

CT-scan collection The dataset consists currently of 48 CT scans, acquired at APHP Armand Trousseau (Paris, France), made publicly available after anonymising (including defacing) and obtaining an ethics approval from the institutional review board for the intended use. The CT scans were acquired using the ... scanner at the hospital Armand Trousseau, Paris, France.

Semi-automatic annotation The annotations consist in voxel-level labels as segmentation masks. For each CT scan, 24 vertebrae are annotated (C1 to C7, T1 to T12, L1 to L5), as well as the anatomy for each vertebrae in 13 labels: vertebra body, intervertebral disk, pedicles (left and right), articular processes (left and right, upper and lower), transverse processes (left and right), lamina (left and right) and spinous process. As medical image annotation is a highly time-consuming task, we reduce the annotation workload by using public datasets and pre-trained deep-learning models found in the literature to produce a preliminary segmentation to be refined into an annotation by the expert annotator.

2 Data annotation

As annotation is long and fastidious, we want to use public datasets and pre-trained models that can be found in the scientific literature to produce a preliminary segmentation to be refined by the expert annotator. Those model can be fine-tuned with the annotations eventually produced by the annotator to produce even more accurate preliminary segmentations for the next batch of CT-scans to annotate.

2.1 Overview of scientific literature

An initial literature review is conducted to identify relevant public datasets and pre-trained models.

2.1.1 Vertebrae annotation

Spine anatomy overview

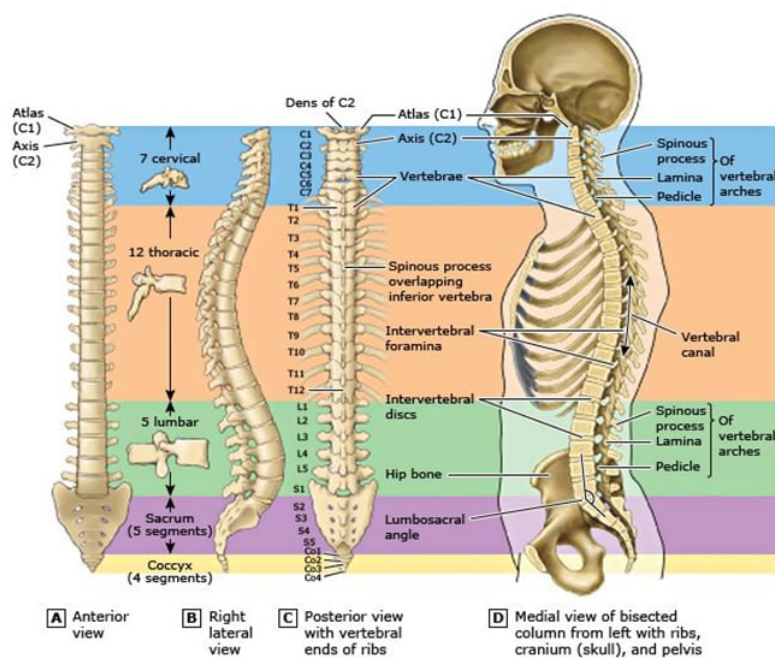


Figure 1: Spinal anatomy. Source: <https://www.drjunkim.com/patient-resources/spinal-anatomy/>

Public datasets Three vertebrae annotation datasets are publicly available : VerSe [1], CT-Spine1k [2] and TotalSegmentator [3]. VerSe is a multi-detector, multi-site, CT spine dataset consisting of 374 scans from 355 patients. CTSpine1k is a multi-detector large-scale dataset, curated from four open sources dataset, totaling 1,005 CT volumes with over 500,000 labeled vertebrae slices

and 11,172 vertebrae belonging to different spinal conditions. TotalSegmentator consists in 1204 CT examinations (from the years 2012, 2016, and 2020 acquired at the University Hospital Basel) with 104 anatomical structures segmented (27 organs, 59 bones, 10 muscles, 8 vessels). A second version of TotalSegmentator contains 1228 CT scans with 117 labels. See Table 1 for comparison between datasets. RODEO (vertebrae) designates the expected RODEO dataset with vertebrae annotations.

Table 1: **Overview of the datasets.** Ticks [✓] in the table refer to that this dataset contains some special cases with sacral lumbarization or lumbar sacralization and we will list their IDs in the open source. ‘Tr\Ts_pu\Ts_pr’ denotes training\test_public\test_private. We exclude the metal-artifact cases due to the difficulty of labeling them.

Dataset name	VerSe20 [✓]	CTSpine1k [✓]	TotalSegmentator	RODEO (vertebrae)
# of patients (n)	300	1005	1228	48
# of scans (n)	319	1005	1228	48
labels of vertebrae	C1-C7, T1-T13, L1-L6	C1-C7, T1-T12, L1-L6	C1-C7, T1-T12, L1-L5, S1, sacrum	C1-C7, T1-T12, L1-L5
# of labeled vertebrae (n)	4141	11172	-	-
# of Tr\Ts_pu\Ts_pr (n)	113\103\103	610\197\198	-	-
# of cervical vertebrae (n)	581	419	-	-
# of thoracic vertebrae (n)	2255	5942	-	-
# of lumbar vertebrae (n)	1305	4712	-	-
# of sacral lumbarization	?	17	-	-
# of lumbar sacralization	?	11	-	-
Mean spacing (mm)	(0.81±1.1, 0.81±1.0, 0.88±1.6)	(0.76±1.1, 0.76±1.1, 1.10±.94)	(1.5±0, 1.5±0, 1.5±0)	(0.55±0.1, 0.55±0.1, 0.62±0.004)
Mean size	(512, 512, 640±197)	(512, 512, 504±155)	-	(512, 512, 1034±130)
Age	56.2 ± 17.6	-	-	< 18 years
Manufacturer	Siemens, GE, Philips, Toshiba	-	-	-
Year	2019, 2020	2008, 2019, 2020	2012, 2016, 2020	2024

Models We found three models: Coarse to Fine Vertebrae Localization and Segmentation with SpatialConfiguration-Net and U-Net [4], nnUNet [5] and TotalSegmentator [3]. [4] was proposed in the VerSe19 challenge and was ranked first on leaderboard. The code was made publicly available, as well the 2020 version for the VerSe20 dataset. The method consists of a 3-staged pipeline. A 3D U-Net regresses a heatmap of the spinal centre line. Individual vertebrae are localized and are identified with the SpatialConfig-Net [6]. Each vertebra is then independently segmented as a binary segmentation. nnUNet is a well known automatic training framework used in several medical segmentation task, and was used in CTSpine1k semi-automatic annotation pipeline. TotalSegmentator is the model trained on the dataset of the same name, using the nnU-Net framework. One of its checkpoint can segment the 24 usual vertebrae with S1 vertebrae and sacrum.

Comparison of the models We compare those three models for the preliminary segmentation: Payer method, nnU-Net and TotalSegmentator. TotalSegmentator checkpoint has been trained on the most CT-scans, but at 1.5 mm resolution. Even upscaled, the predicted segmentations might still be of poor quality, meaning the volumes could appear pixelised and not smooth enough. nnU-Net does not have any vertebrae checkpoint, besides the TotalSegmentator one. We can train on higher-resolution dataset such as VerSe20 and CTSpine1k datasets. Those two methods use the nnU-Net framework that enables automatic fine-tuning. Payer method use a 3-staged hierarchical pipeline and thus seems the most medically aware. It has been designed specifically for the VerSe dataset, which contains deviant anatomic cases such as lumbar sacralization and sacral lumbarization (L6), and T12. A checkpoint trained on VerSe20 is available. To train the model on other datasets, the post-processing following the vertebrae localization needs to be adapted. No fine-tuning code is provided.

2.1.2 Intra-vertebrae annotation

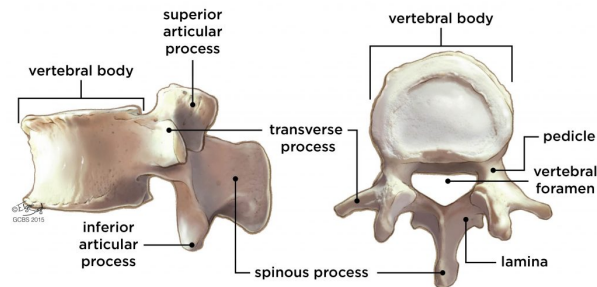


Figure 2: Anatomy of vertebra (here lumbar vertebra). Source: <https://www.goodmancampbell.com/conditions/spine/spine-anatomy/>

Public datasets and checkpoints We found no dataset nor model having intra-vertebrae annotation, except a TotalSegmentator checkpoint that can segment vertebral bodies and intervertebral discs for C1 to S1 (task “vertebrae.body: vertebral body of all vertebrae (without the vertebral arch), intervertebral_discs”). We use this checkpoint to predict a preliminary segmentation, where the structures of the vertebral arc (pedicles, spinous process, etc.) have to be annotated. When a first batch of annotations is done, we can train a nnU-Net model to produce the next batch of preliminary segmentations, this time with the whole anatomy of vertebrae. Another approach can be using a in-context learning segmentation model if the batch of annotations is too small. In-context learning segmentation models, like UniverSeg [7] or IRIS [8], use a context, i.e. examples of images segmented, to segment a target image as the context provided.

2.2 Workflow

To lower the workload, the expert is only tasked with intra-vertebral annotation. For vertebral annotations, we can reconstruct the vertebral volume from the sub-structures of each vertebrae using connected components method.

Vertebral bodies and intervertebral bodies segmentations were produced for the first batch of 48 CT-scans using the TotalSegmentator checkpoint. They constitute the preliminary segmentation for intra-vertebral segmentation to be completed and refined. Depending on the number of annotations produced by the expert on a period of 6 months, an nnU-Net model will be trained or the in-context model IRIS would be used to produce annotations for the rest of the batch.

The vertebral annotations obtained from connected components will be used to evaluate the pre-trained models and identify the model to use in the semi-automatic pipeline.

2.3 Results

We evaluate the best model to use for the semi-automatic annotation pipeline.

2.3.1 Experimental set-up

Vertebrae annotation The performance of the TotalSegmentator checkpoint is evaluated on the CTSpine1k dataset to assess the quality of the predicted segmentations. Concerning Payer model, a checkpoint trained on VerSe20 is available. We adapt the postprocessing after vertebrae localization for CTSpine1k and train a second checkpoint on its training set. We also train two nnU-Net models : one with the training set of VerSe20 and one with the training set of CTSpine1k. We train using a ResEncUNet with the parameters automatically computed by the framework (see Appendix B.1 for details). The cross-validation fold used for VerSe20 are the ones used for Payer checkpoint. For CTSpine1k, we use the one automatically provided by the framework. nnU-Net framework empirically indicates the best configuration for the training dataset using cross-validation. For CTSpine1k, an ensemble of 3D low resolution and 3D cascade configuration appears to be the best configuration and for VerSe20 (see Appendix B.2 for details). However, the inference was time and resource-consuming so we choose the single best configuration, which is 3D full resolution for CTSpine1k and 3D low-resolution for VerSe20. The next results are obtained using a model with those configurations, trained on the whole training set of each dataset.

Intra-vertebrae annotation Vertebral bodies and intervertebral discs segmentations were produced for the first batch of 48 CT-scans using the TotalSegmentator checkpoint as preliminary segmentation for intra-vertebral segmentation. The 48 CT-scans were downsampled to 1.5 mm resolution, feed to the network, and the outputs were upsampled back to their original resolution.

2.3.2 Metrics

To evaluate the quality of segmentation, we use the Dice Score (DS) as computed by nnU-Net, i.e. as the average of the dice scores by vertebra across all cases. We use Hausdorff Distance (HD) and Hausdorff Distance 95 (HD95) as well.

2.3.3 Quantitative results

To choose the best model, we evaluate each trained model in distribution, i.e. on the test set of its training dataset test dataset, and in transfert, i.e. on the test set of another dataset, as presented in Table 2. For nnUNet models, the configurations used for VerSe and for CTSpine1k are 3D low resolution and 3D full resolution. nnUNet has trouble segmenting label 26 (T13), anatomic variant present in VerSe20 dataset. We observe that this vertebra can hinder the segmentation of the neighboring vertebra (T12). We provide results without taking into account this label to have a better seizure of the performances of the models. We observe the same perturbations for the anatomic variant L6, but both datasets have this case. This issue is also present at the vertebrae localization stage in Payer method but is corrected by the postprocessing right after. HD and HD95 are computed resizing predictions to 1mm resolution.

2.3.4 Qualitative results

Vertebrae annotation TotalSegmentator checkpoint provides lower-quality preliminary segmentations, as the quantitative results show. Visually, it is also apparent that the segmentations produced are of lower quality compared to those from other datasets. The nnU-Net trained checkpoints provide better results than Payer method, and would be easier to fine-tune thanks to the automatic

Table 2: **Comparison of pre-trained models** Comparison of Dice score, Hausdorff Distance (HD), and 95th percentile Hausdorff Distance (HD95) across datasets and pre-training configurations.

Model	Pre-training Dataset	VerSe20 (test set 2)			CTSpine1k (test set)		
		Dice	HD (mm)	HD95	Dice	HD (mm)	HD95 (mm)
Payer	VerSe20	80.77	5.05	1.85	70.56	7.75	1.91
	CTSpine1k	65.33	9.76	4.96	65.29	7.96	2.02
nnU-Net	VerSe20	75.57, without label 26: 78.60	7.37	3.26	82.73, without label 26: 86.04	9.00	3.21
	CTSpine1k	70.34, without label 26: 73.16	12.35	4.65	87.00	7.42	2.07
TotalSegmentator	TotalSegmentator dataset	–	–	–	85.70	–	–

framework. The nnU-Net checkpoint trained with CTSpine1k seems to provide the best results. The evaluation in transfert on vertebral annotations will validate those results.

Intra-vertebrae annotation TotalSegmentator predicted segmentations are visually not as high-quality as expected for an annotation dataset. The expert will need to smooth those segmentations. As the structures of the vertebral arc are not annotated, we expect the expert to produce few high-quality annotations in the 6 months and an in-context model will be used to produce the rest of the preliminary segmentations.

3 Conclusion & Perspectives

We are expecting intra-vertebral annotations from the expert to launch the semi-automatic pipeline for both intra-vertebral annotation and vertebral annotation. This dataset is the first scoliosis-focused dataset and the first sub-vertebrae annotation dataset in the literature.

This dataset will be used as augmented view of CT-scan and surgical scene after registration for robotic perception and planning. This will enable safer and faster surgeries for both patients and surgeons, and help trainee surgeons to gain experience and confidence faster.

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A Vertebrae datasets

A.1 Vertebrae datasets illustrations

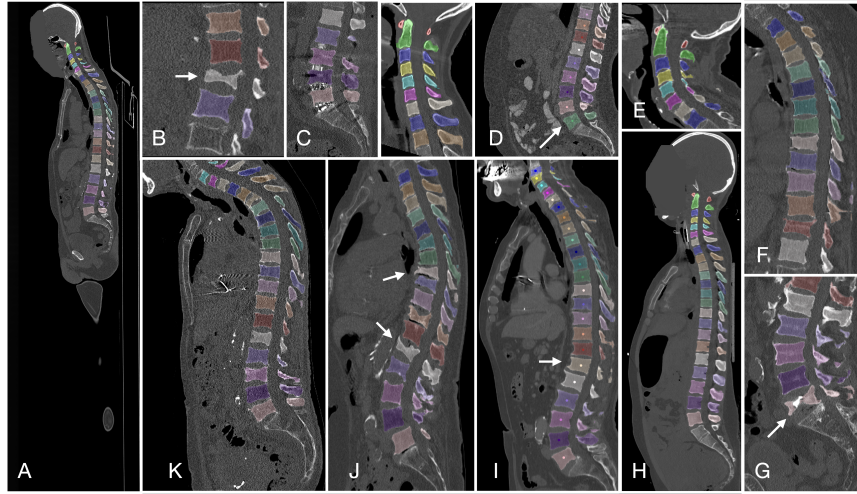


Figure 3: VerSe dataset examples. Observe the variability in data: field-of-view, fractures, transitional vertebrae, etc. Source: <https://github.com/anjany/verse/tree/main>

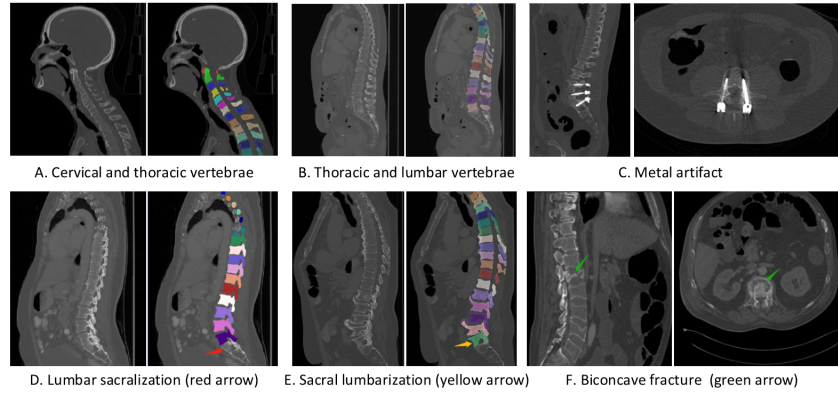


Figure 4: CTSpine1k dataset examples. Source: <https://github.com/MIRACLE-Center/CTSpine1K>

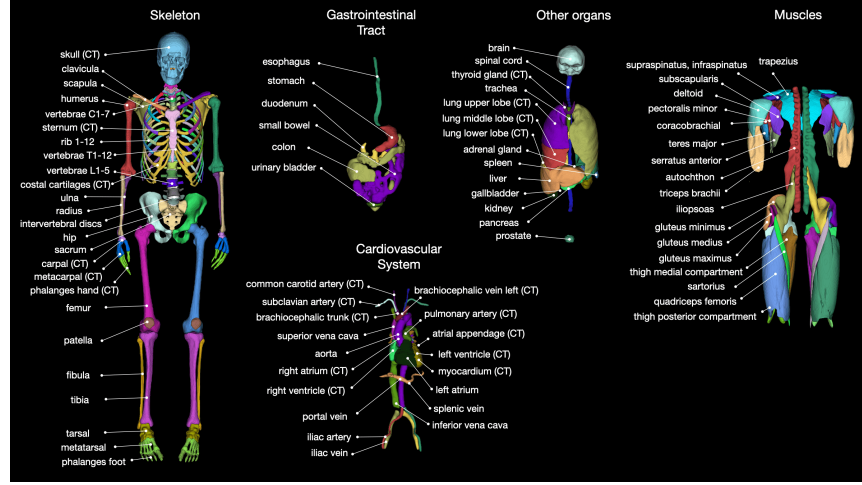


Figure 5: TotalSegmentator: Main classes for CT and MR. Source: <https://github.com/wasserth/TotalSegmentator>

A.2 Vertebrae datasets comparison

Table 3: **Overview the datasets.** Ticks $\checkmark$ in the table refer to that this dataset contains some special cases with sacral lumbarization or lumbar sacralization and we will list their IDs in the open source. ‘Tr\Ts_pu\Ts_pr’ donates training\test_public\test_private. We exclude the metal-artifact cases due to the difficulty of labeling them.

Dataset name	VerSe20 $\checkmark$	COLONOG $\checkmark$	HNSCC-3DCT-RT	MSD T10 $\checkmark$	COVID-19	CTSpinelk $\checkmark$	TotalSegmentator	RODEO (vertebrae)
# of patients (n)	300	784	31	150	40	1005	1228	48
# of scans (n)	319	-	-	-	-	1065	1228	48
labels of vertebrae	C1-C7, T1-T12, L1-L6	-	-	-	-	C1-C7, T1-T12, L1-L6	C1-C7, T1-T12, L1-L5, S1, sacrum	C1-C7, T1-T12, L1-L5
# of labeled vertebrae (n)	4141	4028	443	2105	602	11172	-	-
# of Tr\Ts_pu\Ts_pr (n)	113,103,103	152,152	20,5,6	90,30,30	20,10,10	610,197,198	-	-
# of cervical vertebrae (n)	581	0	217	132	66	419	-	-
# of thoracic vertebrae (n)	2255	4039	226	1229	479	5942	-	-
# of lumbar vertebrae (n)	1305	3911	0	744	57	4712	-	-
# of sacral lumbarization	?	16	0	0	0	17	-	-
# of lumbar sacralization	?	9	0	0	0	11	-	-
Mean spacing (mm)	(0.81 $\pm$ 1.1, 0.81 $\pm$ 1.0, 0.88 $\pm$ 1.6)	(0.75 $\pm$ 0.8, 0.81 $\pm$ 1.6)	(1.09 $\pm$ 1.4, 1.09 $\pm$ 1.4, 2.0)	(0.77 $\pm$ 1.1, 0.77 $\pm$ 1.1, 1.55 $\pm$ 1.22)	(0.79 $\pm$ 0.9, 0.79 $\pm$ 0.9, 4.5 $\pm$ 1.34)	(0.76 $\pm$ 1.1, 0.76 $\pm$ 1.1, 1.10 $\pm$ 0.94)	(1.5, 1.5, 1.5)	(0.8, 0.8, 0.8)
Mean size	(512, 512, 640 $\pm$ 197)	(512, 512, 542 $\pm$ 58)	(512, 512, 202 $\pm$ 10)	(512, 512, 478 $\pm$ 264)	(512, 512, 93 $\pm$ 70)	(512, 512, 504 $\pm$ 155)	-	-
Age	56.2 $\pm$ 17.6	56.2 $\pm$ 6.1 years	64.3 $\pm$ 12.8 years	Unknown	Unknown	-	-	< 18 years
Manufacturer	Siemens, GE, Philips, Toshiba	Siemens, GE, Philips, Toshiba	Siemens	Unknown	Unknown	-	-	-
Year	2019, 2020	2008	2008	2019	2020	2008, 2019, 2020	2012, 2016, 2020	2024

B nnU-Net for vertebrae segmentation: parameters and cross-validation results

B.1 nnU-Net configuration parameters

B.2 Best configuration of nnU-Net using cross-validation

nnU-Net automatically indicates the best configuration or ensemble of configurations by computing the cross-validation dice score for each.

We observe that the 2D configuration has much worse results than other configurations. For CTSpinelk, we will not train the 2D configuration to save resources.

Table 4: **Key nnU-Net Configuration Parameters for VerSe20.**

Configuration	Batch Size	Patch Size (voxels)	Voxel Spacing (mm)	Network Stages
2D	60	(384, 384)	(0.91, 0.98)	7
3D low resolution	2	(256, 160, 160)	(1.34, 1.23, 1.31)	6
3D full resolution	2	(256, 160, 160)	(1.00, 0.91, 0.98)	6
3D cascade full resolution	inherits 3D fullres	–	–	–

Table 5: **Key nnU-Net Configuration Parameters for Dataset200_CTSpine1k.**

Configuration	Batch Size	Patch Size (voxels)	Voxel Spacing (mm)	Network Stages
2D	34	(512, 512)	(0.75, 0.75)	8
3D low resolution	2	(192, 192, 192)	(1.40, 1.32, 1.32)	6
3D full resolution	2	(192, 192, 192)	(0.80, 0.75, 0.75)	6
3D cascade full resolution	inherits 3D fullres	–	–	–

Table 6: **nnU-Net trained with VerSe20 - Dice scores for different nnU-Net configurations and ensembles.** Results computed using 3-fold cross-validation. The best-performing configuration is highlighted in bold.

Configuration	Dice Score
2D	0.5406
3D full resolution	0.7338
3D low resolution	0.7479
3D cascade full resolution	0.7202
Ensemble (2D + 3D full resolution)	0.5905
Ensemble (2D + 3D low resolution)	0.6282
Ensemble (2D + 3D cascade full resolution)	0.6256
Ensemble (3D full resolution + 3D low resolution)	0.7460
Ensemble (3D full resolution + 3D cascade full resolution)	0.7357
Ensemble (3D low resolution + 3D cascade full resolution)	0.7534

Table 7: **nnU-Net trained with CTSpine1k (without liver subset) - Dice scores for different nnU-Net configurations and ensembles.** Results computed using 5-fold cross-validation. The best-performing configuration is highlighted in bold.

Configuration	Dice Score
3D low resolution	0.8644
3D full resolution	0.8567
3D cascade full resolution	0.8731
Ensemble (3D low resolution + 3D full resolution)	0.8728
Ensemble (3D low resolution + 3D cascade full resolution)	0.8738
Ensemble (3D full resolution + 3D cascade full resolution)	0.8746

As inference using an ensemble of models, furthermore one cascade, is time and resource-consuming, we only used the best single model configuration. We do not use either the postprocessing proposed by nnU-Net as the postprocessing does not improve much the performance (for example on VerSe20, 0.7539 vs 0.7534) and is time-consuming.